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Visual Outcomes after Proton Beam Irradiation for Choroidal Melanomas Involving the Fovea

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Purpose: To report visual outcomes in patients undergoing proton beam irradiation of tumors located within 1 disc diameter of the fovea.

Design: Retrospective review.

Participants: Patients with choroidal melanoma involving the fovea treated with proton beam therapy between 1975 and 2009.

Methods: Three hundred fifty-one patients with choroidal melanomas located 1 disc diameter (DD) or less from the fovea and more than 1 DD away from the optic nerve were included in this study. In a subgroup of 203 of the patients with small and medium choroidal melanomas, the effect of a reduced dose of radiation, 50 Gy (relative biological effectiveness [RBE]) versus 70 Gy (RBE), on visual outcomes was analyzed. The Kaplan-Meier method and Cox regression analysis were performed to calculate cumulative rates of vision loss and to assess risk factors for vision loss, respectively.

Main Outcome Measures: Visual acuity and radiation complications, which included radiation maculopathy, papillopathy, retinal detachment, and rubeosis, were assessed.

Results: Three hundred fifty-one patients were included in this study with a mean follow-up time of 68.7 months. More than one-third of patients (35.5%) retained 20/200 or better vision 5 years after proton beam irradiation. For those patients with a baseline visual acuity of 20/40 or better, 16.2% of patients retained this level of vision 5 years after proton beam irradiation. Tumor height less than 5 mm and baseline visual acuity 20/40 or better were associated significantly with a better visual outcome ($P < 0.001$). More than two-thirds (70.4%) of patients receiving 50 Gy (RBE) and nearly half (45.1%) of patients receiving 70 Gy (RBE) retained 20/200 or better vision 5 years after treatment, but this difference was not significant. Approximately 20% of patients with these smaller macular tumors retained 20/40 vision or better 5 years after irradiation.

Conclusions: The results of this retrospective analysis demonstrate that despite receiving a full dose of radiation to the fovea, many patients with choroidal melanoma with foveal involvement maintain useful vision. A radiation dose reduction from 70 to 50 Gy (RBE) did not seem to increase the proportion of patients who retain usable vision. *Ophthalmology 2016;123:369-377 © 2016 by the American Academy of Ophthalmology.*

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Choroidal melanoma is the most common primary intraocular malignancy among adult patients and is one of the few potentially fatal ocular diagnoses. Treatment options for choroidal melanomas include enucleation, laser photocoagulation, transpupillary thermotherapy, and irradiation. Over the past few decades, radiotherapy has become used more widely in the care of these tumors.[1–4] Achieving local tumor control, while preserving the eye and maintaining some useful vision, is the primary goal of radiotherapy in the management of uveal melanoma. Two major radiotherapy methods currently are used: external beam radiation therapy, most commonly with protons, and radioactive plaques, which are placed over the sclera in the area of the tumor.[2,4–6] As charged particles, protons can provide highly localized radiation dose distributions, and by doing so may reduce ocular morbidity.[3]

Radiation maculopathy and papillopathy are common sequelae of radiation treatment and are the predominant causes of visual loss in patients after radiotherapy. Radiation maculopathy is the main contributor to vision loss in patients who have tumors with foveal involvement.[7] The clinical manifestations of radiation maculopathy include capillary closure, telangiectasia, microaneurysm formation, hemorrhages, exudates, macular edema, and nerve fiber layer infarctions.[8] Previous studies have shown that the risk of radiation maculopathy and papillopathy are determined primarily by the radiation dose to the macula and optic nerve, respectively.[9,10] Additionally, underlying

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